

Building association matrices

CONCEPT AND STRATEGIC VALUE

An Association Matrix is a structured table that maps **Events** (rows) to **Indicators** (columns), transforming narrative ideas into quantifiable modeling objects.

THE STRATEGIC BRIDGE

It acts as the essential link between a chronological timeline of events and the specific behavioral shifts in a forecast model.

Core Purpose: Prevents arbitrary assumptions by making impact logic transparent, reviewable, and auditable.

KEY OUTCOME

Ensures high explainability, where every forecast change is directly traceable to specific cells within the matrix.

STRUCTURING ROWS AND COLUMNS

Rows: Events

Major Milestones: Policy reforms, product launches, and infrastructure upgrades.

Requirements: Fixed dates and clear start points for every entry.

Unique IDs: Assign stable identifiers (e.g., EVT_001) for linking and tracking.

Columns: Indicators

Headline Outcomes: Focus on primary metrics like Access and Usage indicators.

Supporting Drivers: Include coverage, affordability, and adoption metrics.

Consistency: Uniform units and definitions across the entire timeline.

ANATOMY OF A MATRIX CELL

01 DIRECTION: Increase (+) or Decrease (-)

02 MAGNITUDE: Numeric estimate or tier-based scale

03 LAG: Time delay before effect manifests

04 DURATION: Shock, gradual lift, or temporary shift

05 CONFIDENCE: Assessment of evidence strength

PRACTICAL POPULATION WORKFLOW

01

SHORTLISTING

Identify realistic event-indicator links to avoid noise.

02

MECHANISM LOGIC

Define the specific causal channel linking event to outcome.

03

EVIDENCE-BASED VALIDATION

Use data breaks, trend shifts, or peer comparisons.

04

DOCUMENTATION

Record all assumptions to ensure transparency and auditability.

EVOLUTION: BINARY TO WEIGHTED

BINARY MATRIX

A simple **1/0 mapping** used to identify connectivity.
Ideal for high-level overviews and checking event-indicator relationships.

WEIGHTED MATRIX

Stores specific **impact estimates**. Enables direct integration into mathematical models and simulation engines.

$$\text{Forecast} = \text{Baseline Trend} + \Sigma(\text{Active Event Effects})$$

DATA INTEGRATION & IMPACT LINKS

The matrix is a dynamic view of your underlying dataset's **Impact Links**.

IMPACT SHEETS

The "long-format" version of the matrix. Each row represents a single link: **Event** → **Indicator** with metadata.

CONSISTENCY

Enforces standard code lists for pillars and confidence levels, ensuring high data integrity.

DYNAMIC UPDATING

Pivot impact records to generate the matrix automatically. Editing a link row instantly updates the modeling view.

EFFICIENCY

Direct linking ensures scalability and allows for automated validation of impact logic.

FORECASTING & SCENARIO MODELING

BASELINE VS. EVENT-AWARE

Compare standard trends against forecasts that incorporate specific event impacts applied after their respective lags.

SENSITIVITY TESTING

Vary magnitude and confidence values to assess forecast risk and identify high-impact assumptions.

SCENARIO TOGGLING

Enable or disable specific event rows to test hypothetical futures, allowing for rapid "what-if" analysis.

STRATEGIC INSIGHTS

Generate explainability outputs that identify the top contributing events for any given indicator forecast.

QUALITY STANDARDS FOR SUCCESS

SELECTIVE DENSITY

Most cells should remain empty. Focus only on plausible, significant links to prevent model noise.

COMPLETENESS

Every filled cell must include direction, magnitude, lag, and confidence. No partial links.

LOGICAL ALIGNMENT

Impacts must match the mechanism. Pricing affects affordability first, then usage trends.

USABILITY

The matrix must be exportable in standard formats (CSV/JSON) for integration with modeling tools.

CONCLUSION & NEXT STEPS

Association matrices turn qualitative insights into quantitative power, bridging narrative and model.

They provide the transparency needed for robust, explainable forecasting in complex environments.

By structuring impacts through direction, magnitude, and lag, they enable precise scenario modeling.

Thank you!